Supplementary Material: When Compact Is Not Enough

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This supplement contains the temporal-alignment audit figure, the sixteen prespecified subgroup estimates and the per-seed locked-test results supporting the main paper. All values are measured under protocol hash `380269a4a6db7334`; subgroup contrasts are exploratory and unadjusted for multiplicity.

1 TEMPORAL ALIGNMENT AUDIT

Figure 1 shows the synthetic-impulse audit that every cache must pass before conversion.

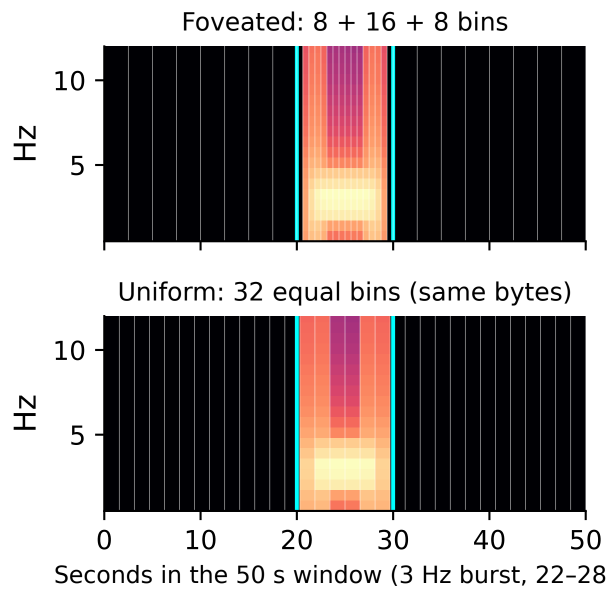

Fig. 1. A synthetic 3 Hz burst confined to seconds 22–28 passed through the real encoder. Foveated bins are drawn with their true widths; the labelled centre [20, 30) s is marked in cyan; the uniform control spends the same bytes with equal bins.

2 IMPLEMENTATION AND REPRODUCIBILITY DETAILS

Cache conversion groups windows by recording, reads each Parquet file once, writes float16 values and packed masks into pre-allocated memory-mapped shards of 2,048 rows through `.partial` files renamed atomically only after both views exist for every row, and checksums each shard into a manifest; pinning BLAS to one thread per worker was necessary because multi-threaded matrix products made the context pass two orders of magnitude slower. **Training** runs under a supervisor that samples CUDA allocation, process-tree memory and system availability every fifty steps and aborts at declared stops; a `flock`-based lock forbids concurrent GPU jobs, and every job appends its wall time to the ledger whether it passed or failed. **Roles** are enforced in code: a development-role loader raises on any request for the test partition, and the evaluator role logs each test access against the protocol hash. **Release** is an allowlisted export with identifier and secret scans of the working tree and of every reachable Git object; notebooks are published with aggregate-only outputs, per-row predictions and weights stay private, and public tables suppress cells with fewer than ten patients. The repository ships 170 synthetic CPU tests, eight executed notebooks, the protocol document and every figure with a provenance sidecar recording the hashes it was drawn from.

3 DESIGN GUIDANCE FOR PRACTITIONERS

Cap and count the evidence so that “compact” is measurable; freeze the schedule rather than the checkpoint, because checkpoint selection on a tuning set overstates what a complete refit delivers; choose the comparator on data; calibrate on a partition that never trains weights and report the temperature; and audit evidence by deletion and preprocessing by perturbation, because the two audits disagreed about which model is safer.

4 DATA GOVERNANCE

The release contains no EEG samples, spectrogram values, identifiers, split membership tables, per-window predictions or trained weights; each is reconstructible by an authorised holder of the source data from the published hashes and code. These are data-minimisation measures, not an anonymisation guarantee.

5 DEVELOPMENT PHASE (SEED 101)

Table 1 lists the eight named GPU configurations of the development phase, the two CPU baselines, and the two permitted common-shorter-schedule pilots that fixed the final 3-epoch schedule.

Table 1. Development phase (tune partition, seed 101). GPU configurations 1–8 were the named cap; P_3 and $B3_3$ are the permitted common-shorter-schedule pilot, which was frozen for the final refits.

ID	Configuration	Params	Tune KL	Schedule (best ep.)
B0	vote prior (CPU)	0	1.410	–
B1	band-power regression (CPU)	450	1.241	LBFGS
B2	uniform bins, fixed fusion	147,524	0.986	12 ep (2)
A1	B2 + foveation	147,524	1.001	12 ep (2)
A2	A1 + learned gate	155,877	0.990	12 ep (1)
P	A2 + disagreement aux	164,134	0.983	12 ep (2)
P+MSF	P + multi-scale block	208,070	0.988	12 ep (1)
B3	MobileNetV3-Small	1,524,150	0.896	12 ep (3)
P_3	pilot 7	164,134	0.9270	3 ep, complete
$B3_3$	pilot 8	1,524,150	0.9266	3 ep, complete

6 CALIBRATION AND REVIEW PRIORITISATION

7 PAIRED WITHIN-SEED CONTRASTS OF THE POST-LOCK ABLATIONS

8 PRESPECIFIED SUBGROUP ESTIMATES

Table 3 lists the sixteen strata frozen in the protocol document.

9 PER-SEED LOCKED-TEST RESULTS

Table 4 gives the per-seed values behind the three-seed means of the main paper.

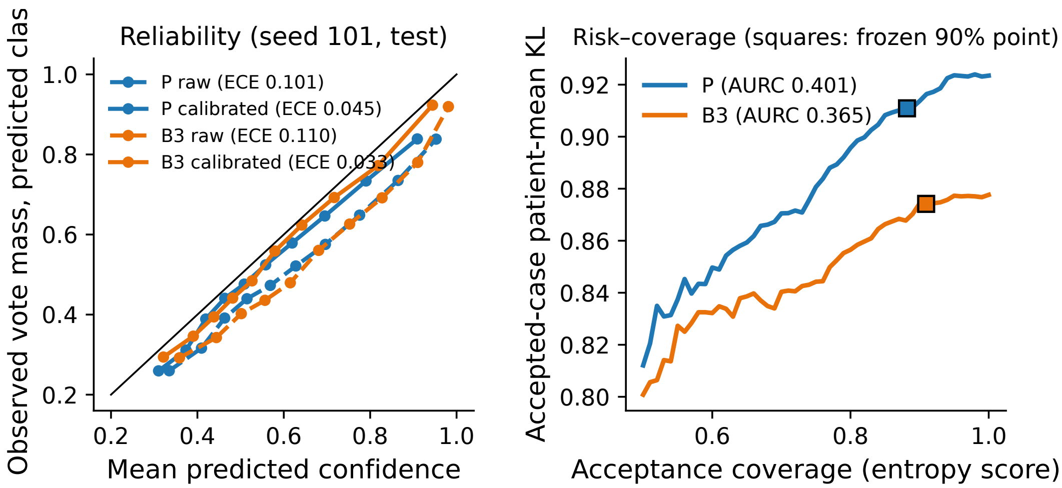

Fig. 2. Left: reliability of P and B3 before and after temperature scaling (seed 101, locked test). Right: KL risk–coverage with the frozen entropy score; squares mark the operating points whose thresholds were fixed on Calibration-P.

Table 2. Paired Δ tune patient-mean KL (mean ± SD over seeds 101/202/303; negative = first term better) under the frozen 3-epoch schedule.

Contrast	Δ
Foveation (A1 − B2)	-0.001 ± 0.035
Learned gate (A2 − A1)	+0.011 ± 0.036
Disagreement loss (P − A2)	-0.000 ± 0.056
All three (P − B2)	+0.010 ± 0.056
ImageNet pretraining (B3 − B3-scratch)	-0.250 ± 0.086
Full vs half width (B3 − B3-half)	-0.193 ± 0.040
B3-scratch − P	+0.160 ± 0.060
B3-half − P	+0.102 ± 0.051

Table 3. Paired calibrated difference in patient-mean KL (P − B3, three seeds) within the sixteen strata frozen before the test; exploratory, unadjusted. Positive favours the comparator.

Stratum	Rows	Patients	KL P	KL B3	Δ [95% CI]
1 vote	510	28	0.939	0.817	+0.123 [−0.032, +0.289] (low support)
2–4 votes	11,990	353	1.057	1.000	+0.057 [+0.013, +0.099]
5–9 votes	315	56	0.491	0.473	+0.018 [−0.043, +0.074]
≥10 votes	8,490	239	0.744	0.666	+0.079 [+0.027, +0.129]
Entropy zero (unanimous)	10,877	357	1.046	0.973	+0.073 [+0.028, +0.116]
Entropy middle tertile	2,380	168	0.736	0.643	+0.094 [+0.051, +0.141]
Entropy upper tertile	8,048	250	0.815	0.796	+0.019 [−0.033, +0.071]
Valid cells $< 95\%$	1,523	134	1.036	0.917	+0.119 [+0.029, +0.206]
Valid cells $\geq 95\%$	19,782	381	0.917	0.853	+0.064 [+0.030, +0.096]
Majority Seizure	4,363	191	0.983	0.928	+0.056 [−0.007, +0.114]
Majority LPD	2,485	52	1.171	0.992	+0.179 [+0.046, +0.308]
Majority GPD	3,143	65	0.983	0.901	+0.082 [−0.062, +0.235]
Majority LRDA	3,137	93	1.580	1.609	−0.028 [−0.154, +0.093]
Majority GRDA	3,608	140	0.979	0.940	+0.039 [−0.030, +0.108]
Majority Other	3,746	239	0.777	0.709	+0.068 [+0.024, +0.108]
Tied majority	823	97	0.760	0.718	+0.042 [−0.021, +0.106]

Table 4. Per-seed locked-test results (temperature-calibrated) and per-class AUROC on unique-majority windows. T = temperature fitted on Calibration-T; AURC = area under the KL risk–coverage curve (coverage 0.5–1.0) with the frozen entropy score.

Method	Seed	T	Patient KL	Row KL	Bal. acc.	AURC	Seizure	LPD	GPD	LRDA	GRDA	Other
P	101	1.221	0.9235	0.8886	0.537	0.401	0.885	0.873	0.925	0.801	0.900	0.782
P	202	1.386	0.9546	0.9142	0.508	0.406	0.887	0.859	0.926	0.788	0.903	0.767
P	303	1.210	0.9244	0.9140	0.523	0.415	0.889	0.847	0.913	0.808	0.895	0.779
B3	101	1.333	0.8776	0.8085	0.572	0.365	0.915	0.903	0.932	0.813	0.915	0.819
B3	202	1.279	0.8564	0.8018	0.571	0.366	0.916	0.910	0.915	0.837	0.923	0.805
B3	303	1.299	0.8715	0.8094	0.585	0.372	0.917	0.909	0.918	0.845	0.915	0.808
P+MSF	101	1.419	0.9478	0.9331	0.491	0.427	0.874	0.847	0.903	0.772	0.906	0.768
P+MSF	202	1.429	0.9568	0.9112	0.520	0.406	0.887	0.868	0.915	0.780	0.897	0.762
P+MSF	303	1.349	0.9696	0.9299	0.509	0.419	0.880	0.850	0.913	0.786	0.888	0.781